COMPLETE PATENT REPORT

A System and Method for Intelligent Event-Based Vehicle Data Acquisition and Transmission Using OBD, CAN, and LIN Networks

1. Title of the Invention

A System and Method for Hardware-Based Selective Vehicle Data Extraction and Event-Driven Transmission Using Multi-Bus Automotive Communication Networks

2. Field of the Invention

The present invention relates to automotive electronics, embedded systems, and intelligent vehicle telematics. More particularly, it relates to a hardware-implemented system for acquiring, processing, and transmitting vehicle data from multiple communication networks including OBD-II, Controller Area Network (CAN), and Local Interconnect Network (LIN), using selective signal extraction and event-driven communication.

3. Background of the Invention

Modern vehicles employ multiple Electronic Control Units (ECUs) interconnected via communication protocols such as CAN and LIN. Standard diagnostic access through OBD-II provides limited information based on predefined parameter identifiers.

Existing vehicle telematics systems exhibit several limitations:

- Continuous streaming of raw vehicle data results in high bandwidth consumption
- Dependence on predefined parameters restricts access to body control data
- Transmission of redundant data increases storage and processing costs
- Lack of intelligent edge-level filtering
- Limited compatibility across different vehicle architectures

Existing solutions from organizations such as Bosch and Geotab primarily focus on data acquisition and transmission rather than selective extraction and event-driven processing at the hardware level.

4. Objects of the Invention

The primary objectives of the invention are:

1. To provide a system capable of acquiring vehicle data from multiple communication protocols.
2. To implement hardware-level processing for selective extraction of relevant signals.
3. To reduce redundant data transmission by transmitting only event-based information.
4. To provide a vehicle-independent detection mechanism adaptable across multiple vehicle models.
5. To optimize communication bandwidth and storage utilization.

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5. Summary of the Invention

The present invention provides a hardware-implemented system comprising a microcontroller, multi-bus communication interfaces, memory, and a wireless communication module.

The system performs:

- Acquisition of raw data from CAN, LIN, and OBD networks
- Storage of baseline signal patterns
- Bit-level differential analysis between incoming data and baseline
- Detection of state transitions corresponding to vehicle events
- Encoding of detected events into compact packets
- Transmission of only event-based data to a remote server

This approach eliminates redundant transmission of unchanged data and significantly improves system efficiency.

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6. Brief Description of Drawings

Figure 1: System Architecture

Figure 2: Multi-Bus Data Acquisition Flow

Figure 3: Bit-Level Event Detection Mechanism

Figure 4: Event-Based Transmission Flow

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7. Detailed Description of the Invention

7.1 System Architecture

The system comprises:

- **Microcontroller Unit (MCU):** Embedded processor configured for real-time data processing
- **CAN Interface:** For high-speed vehicle communication
- **LIN Interface:** For low-speed subsystem communication
- **OBD Interface:** For standardized diagnostic access
- **Memory Unit:** For storing baseline and mapping data
- **Wireless Module:** GSM/LTE-based communication
- **Power Interface:** Connected to vehicle power supply

7.2 Core Working Mechanism

Step 1: Initialization

All communication interfaces and modules are initialized.

Step 2: Baseline Pattern Storage

The system records initial vehicle data and stores it as baseline patterns.

Step 3: Continuous Monitoring

Incoming frames from CAN/LIN/OBD are continuously captured.

Step 4: Bit-Level Differential Processing

The system performs comparison using:

- XOR-based detection
- Bit masking
- Threshold evaluation

Example Implementation:

```
difference = (current_frame XOR baseline_frame) AND mask
```

If `difference ≠ 0`, a state change is detected.

Step 5: Event Identification

Detected differences are mapped to physical vehicle events such as:

- Door open/close

- Ignition ON/OFF
- Window operation

Step 6: Event Encoding

Each detected event is encoded into a compact packet containing:

- Event ID
- Timestamp
- Signal identifier

Step 7: Event-Based Transmission

Only encoded event packets are transmitted via the wireless module.

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8. Technical Effect

The invention achieves the following technical effects:

- Reduction in communication bandwidth through suppression of redundant data
- Reduction in processing load on remote servers
- Real-time event detection at the edge device
- Improved system scalability across vehicle models
- Efficient utilization of IoT communication networks

These effects establish that the invention is a technical solution and not merely an algorithm, thereby addressing objections under Section 3(k) of the Indian Patent Act.

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9. Claims

9.1 Independent Claim

1. A system for vehicle data acquisition and transmission comprising:
    - a microcontroller-based processing unit;
    - a plurality of communication interfaces configured to receive vehicle data from CAN, LIN, and OBD networks;
    - a memory configured to store baseline signal patterns;
    - a signal processing module configured to:
        - perform bit-level comparison between incoming data and baseline patterns,
        - detect state transitions using differential analysis,

        - identify signal variations corresponding to vehicle events;
    - an event encoding module configured to generate compact event packets;
    - a wireless communication module configured to transmit only said event packets;

  wherein the system suppresses transmission of unchanged data and transmits only state-transition-based information.

9.2 Dependent Claims

2. The system wherein the differential analysis is performed using XOR and masking operations.
3. The system wherein baseline patterns are dynamically updated.
4. The system wherein event packets include timestamp and signal identifiers.
5. The system wherein multiple vehicle communication buses are processed simultaneously.
6. The system wherein transmission occurs only upon detection of state change.
7. The system wherein the system operates independently of predefined parameter identifiers.

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10. Industrial Applicability

The invention is applicable in:

- Fleet management systems
- Automotive diagnostics
- Vehicle security systems
- Insurance telematics
- Smart aftermarket vehicle devices

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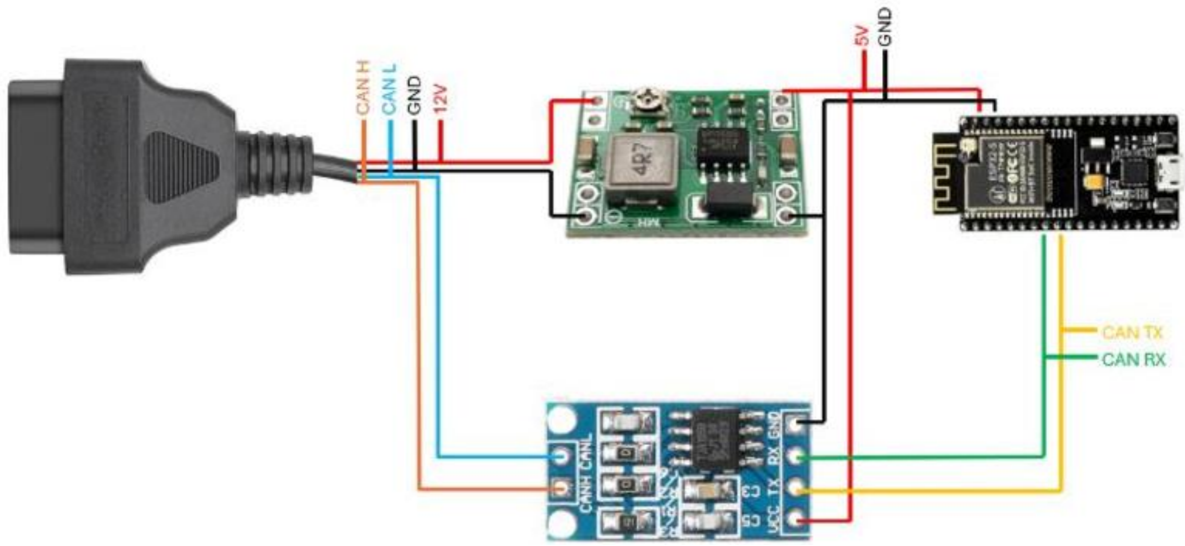
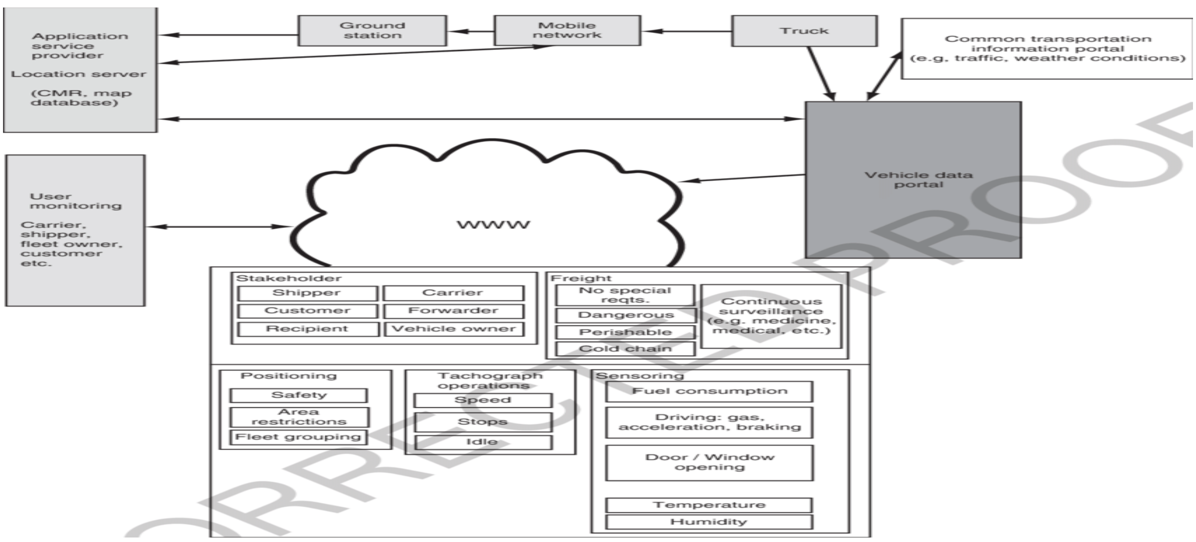
11. Abstract

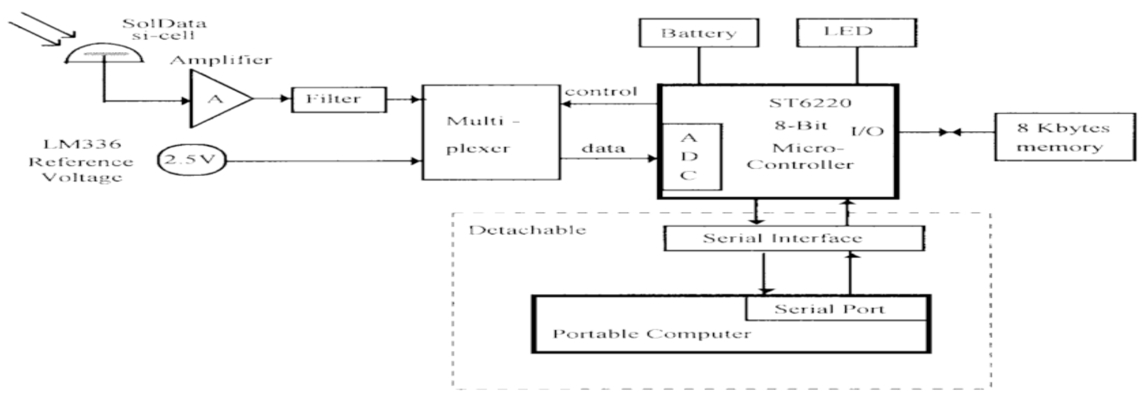
The present invention discloses a system and method for hardware-based selective extraction and event-driven transmission of vehicle data using OBD, CAN, and LIN networks. The system employs a microcontroller to perform bit-level differential analysis between incoming data and stored baseline patterns to detect state transitions corresponding to vehicle events. Only event-based data is transmitted, thereby reducing communication bandwidth and improving system efficiency. The invention is adaptable across multiple vehicle platforms and provides a scalable solution for intelligent vehicle telematics.

DRAWINGS (FINAL SUBMISSION GUIDE)

Figure 1: System Architecture







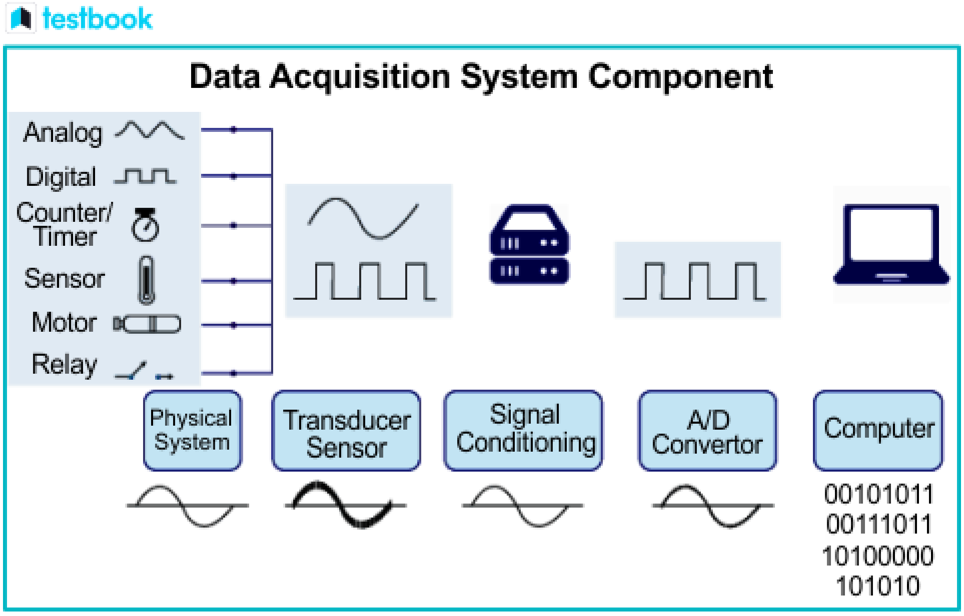
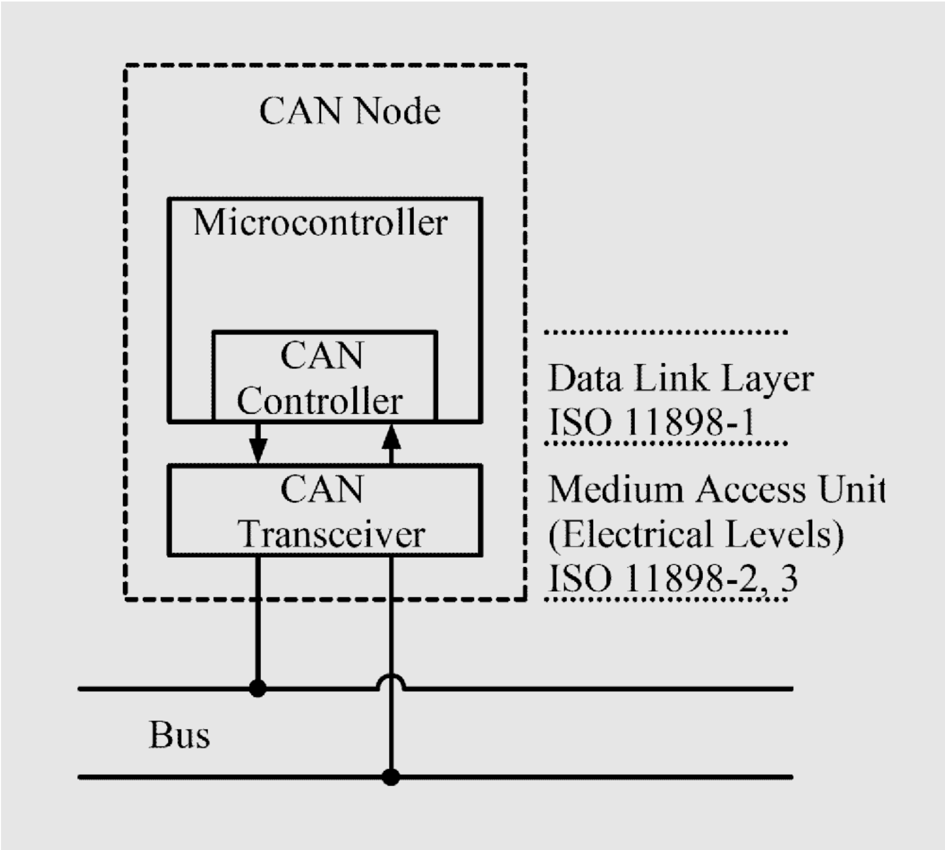
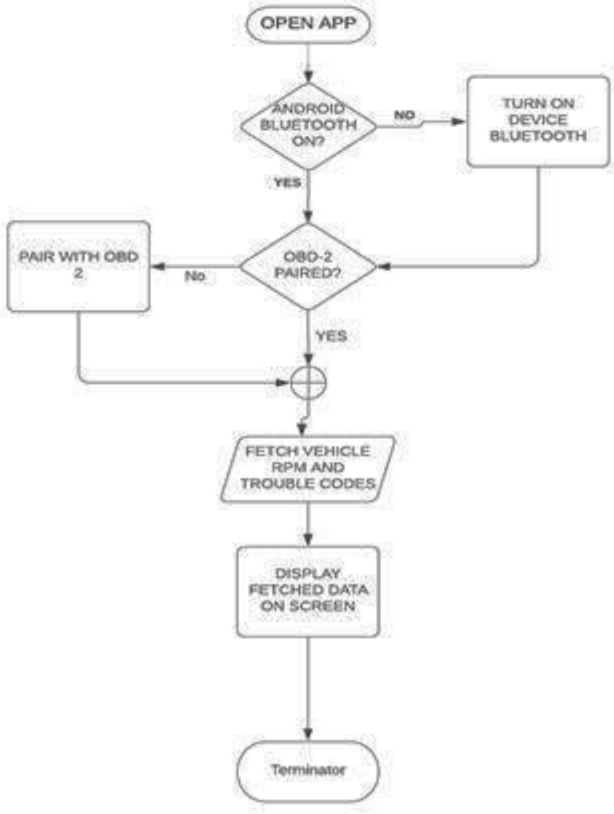
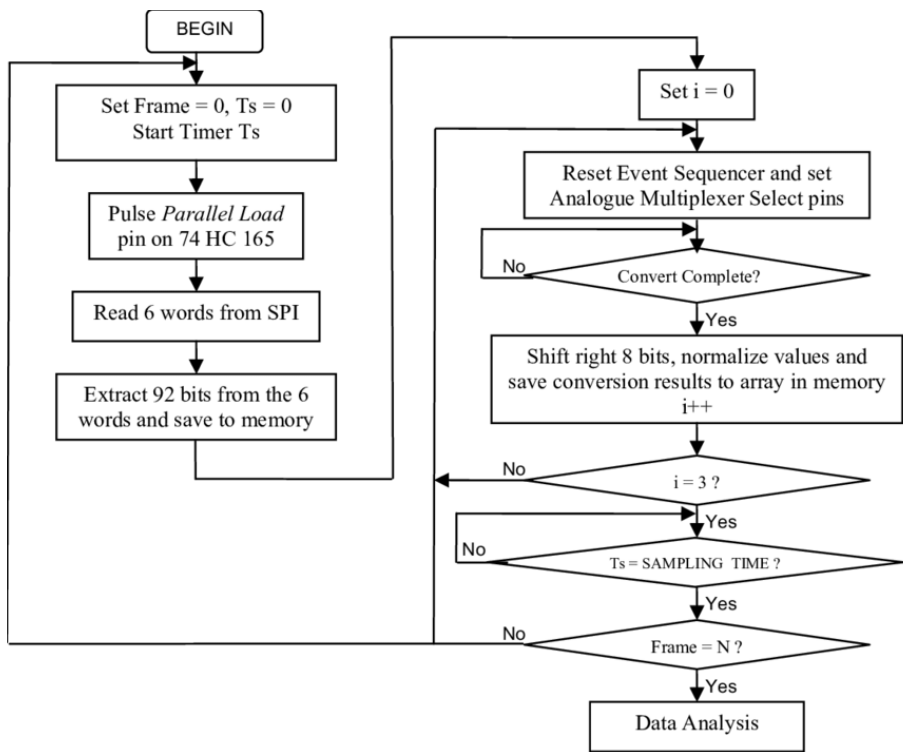


Figure 2: Data Acquisition Flow




OPEN APP
ANDROID BLUETOOTH ON?
NO
TURN ON DEVICE BLUETOOTH
YES
PAIR WITH OBD 2
No
OBD-2 PAIRED?
YES
FETCH VEHICLE RPM AND TROUBLE CODES
DISPLAY FETCHED DATA ON SCREEN
Terminator


BEGIN
Set Frame = 0, Ts = 0
Start Timer Ts
Pulse Parallel Load pin on 74 HC 165
Read 6 words from SPI
Extract 92 bits from the 6 words and save to memory
Set i = 0
Reset Event Sequencer and set Analogue Multiplexer Select pins
No
Convert Complete?
Yes
Shift right 8 bits, normalize values and save conversion results to array in memory
i++
No
i = 3 ?
Yes
No
Ts = SAMPLING TIME ?
Yes
No
Frame = N ?
Yes
Data Analysis

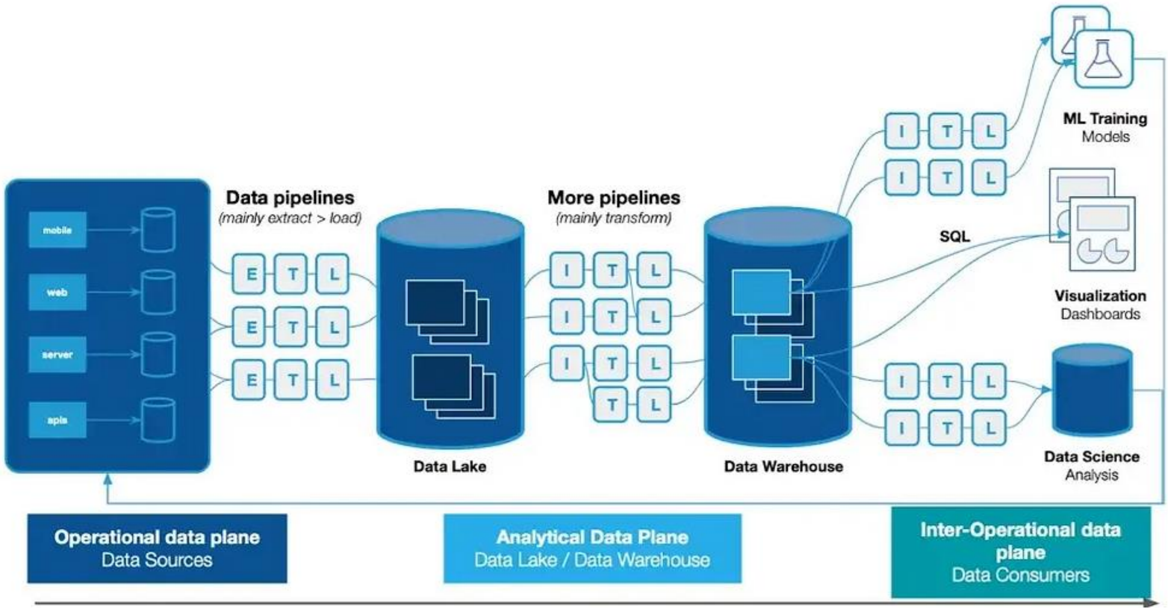
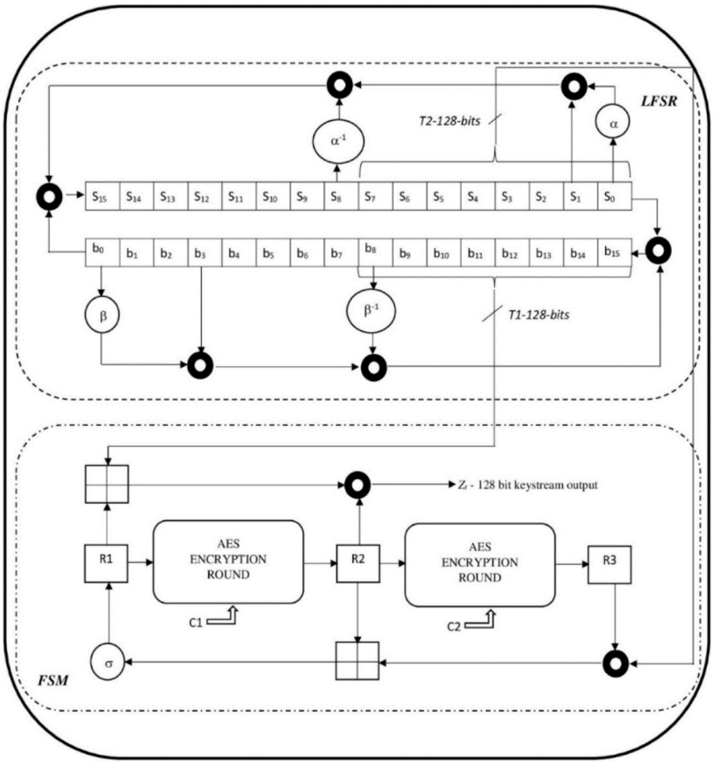
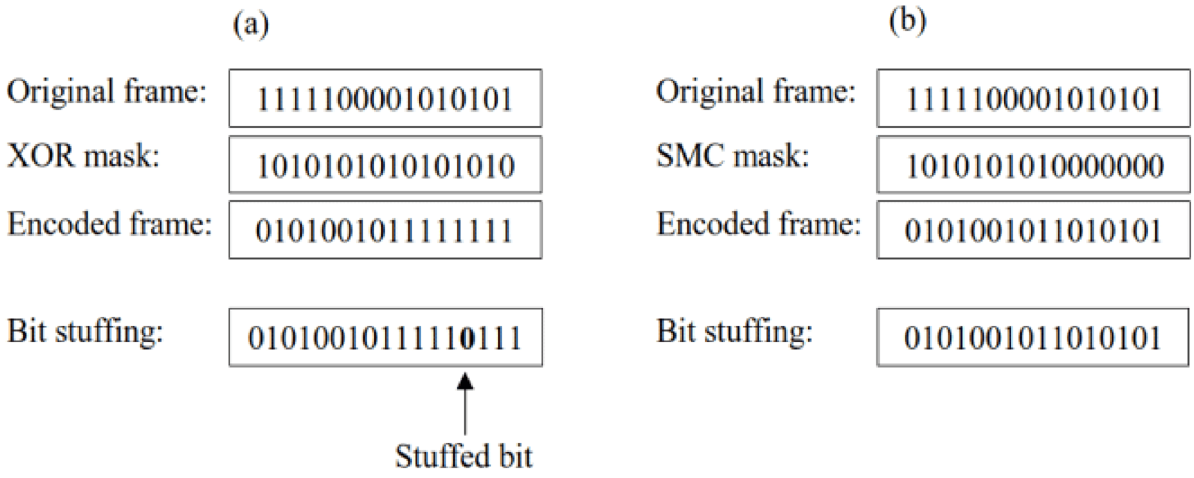


Figure 3: Bit-Level Detection





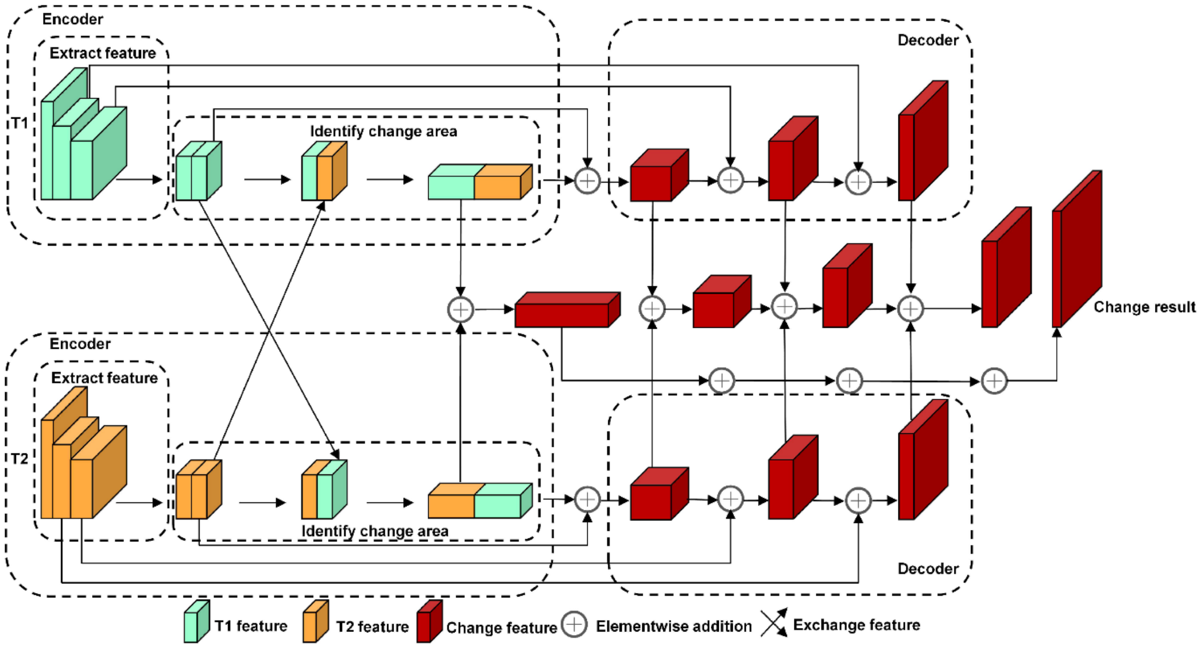
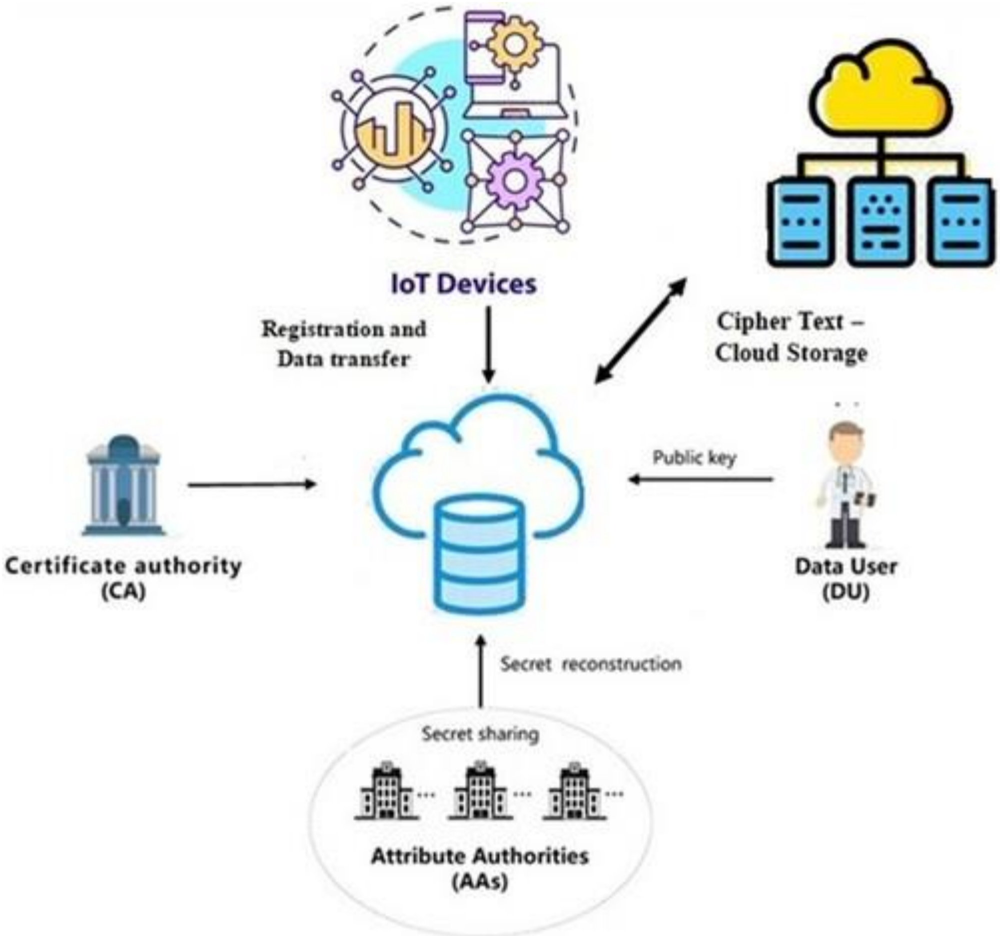
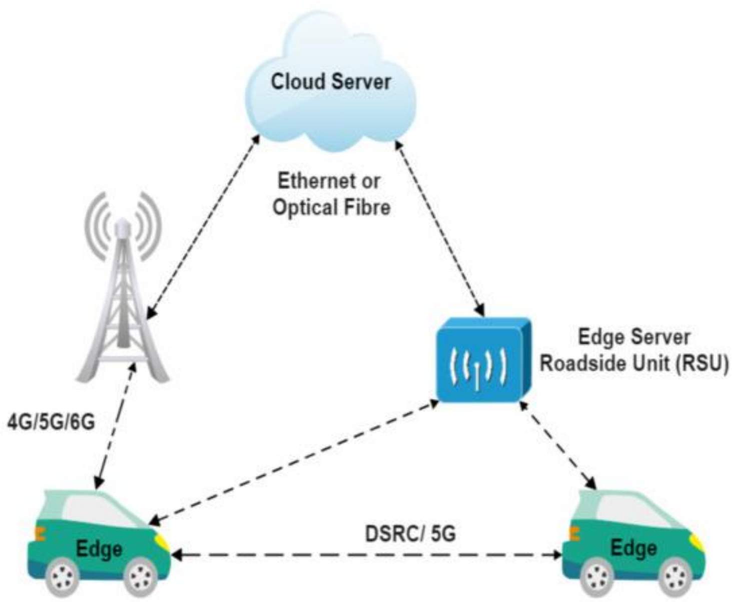


Figure 4: Event Transmission


IoT Devices
Registration and Data transfer
Cipher Text – Cloud Storage
Public key
Certificate authority (CA)
Data User (DU)
Secret reconstruction
Secret sharing
Attribute Authorities (AAs)


Cloud Server
Ethernet or Optical Fibre
Edge Server Roadside Unit (RSU)
4G/5G/6G
Edge
DSRC/ 5G
Edge

Device Images :

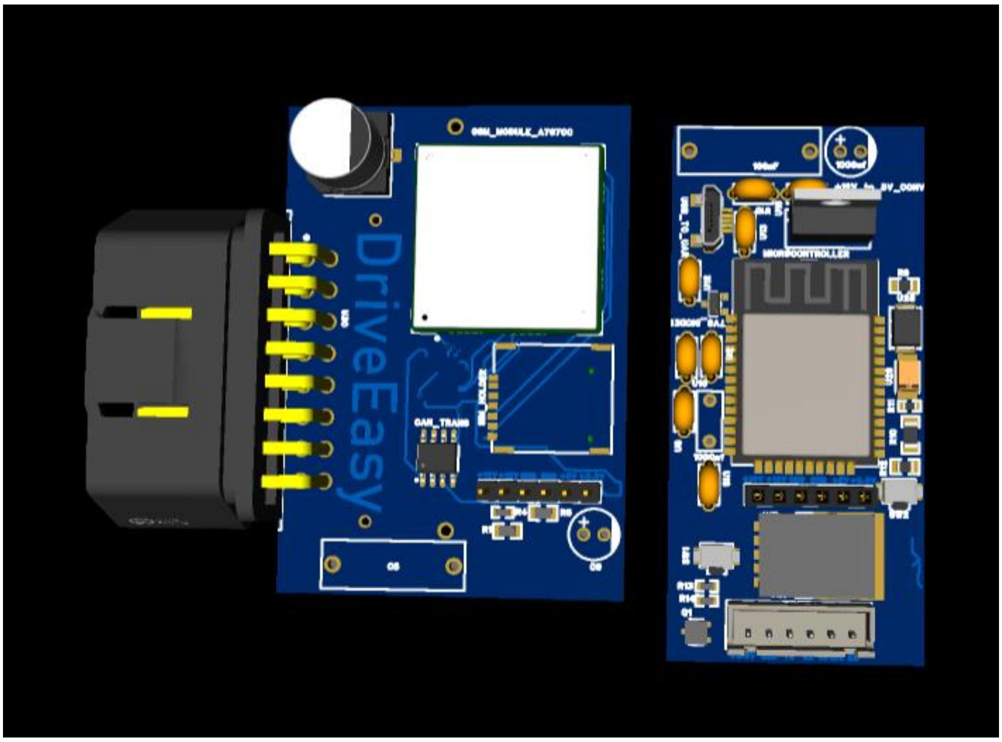


Figure : Top View

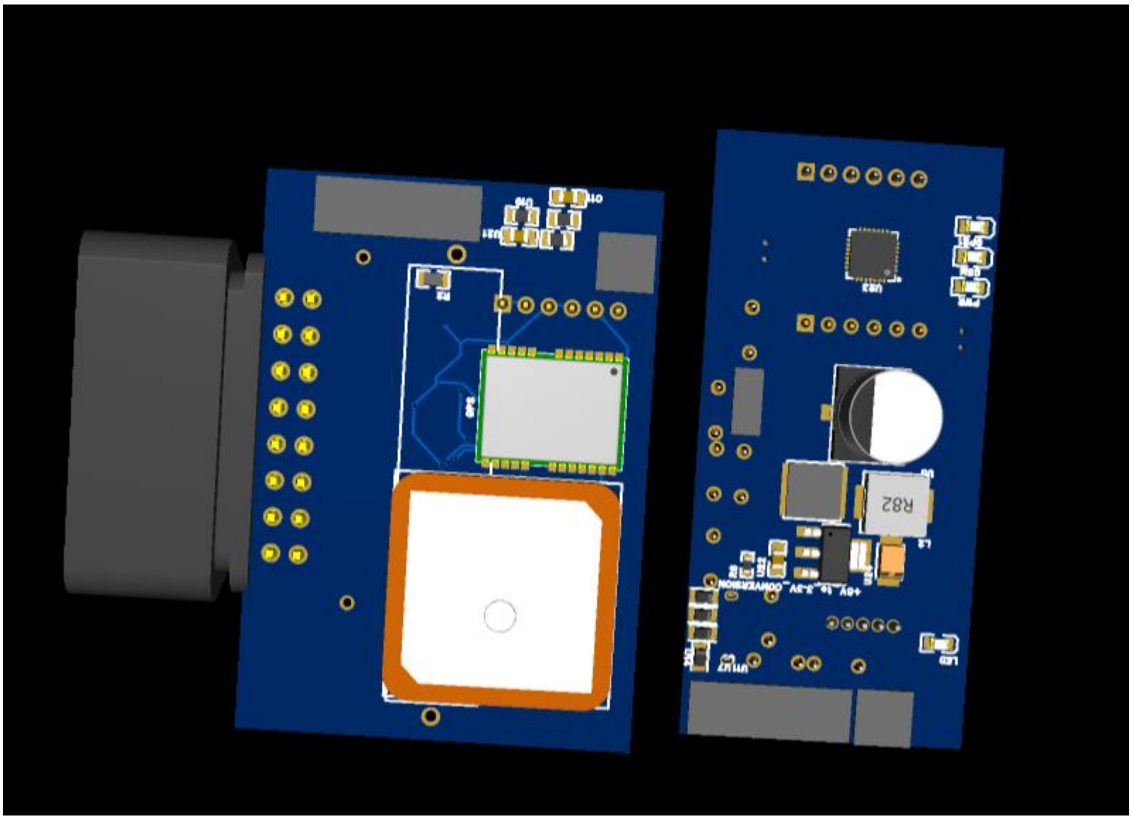


Figure : Bottom View

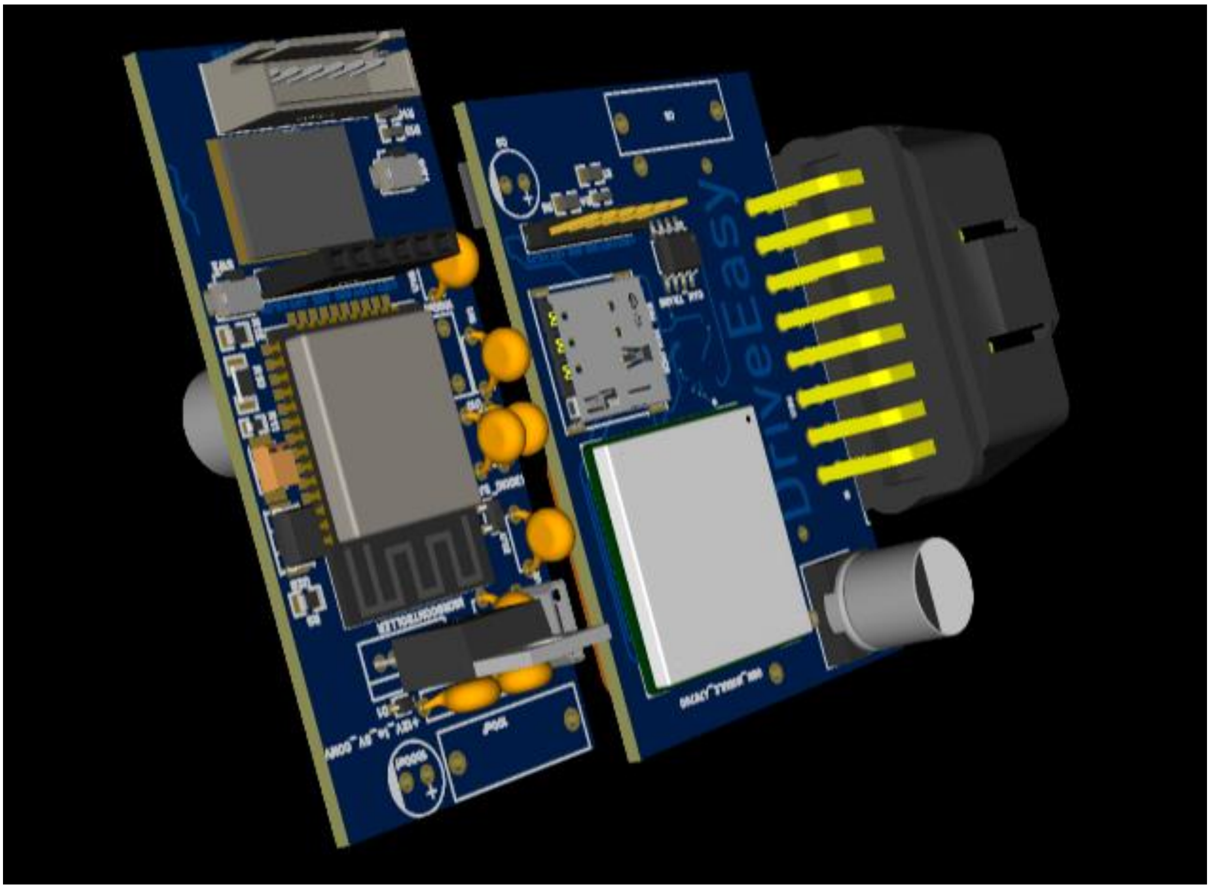


Figure : Left View

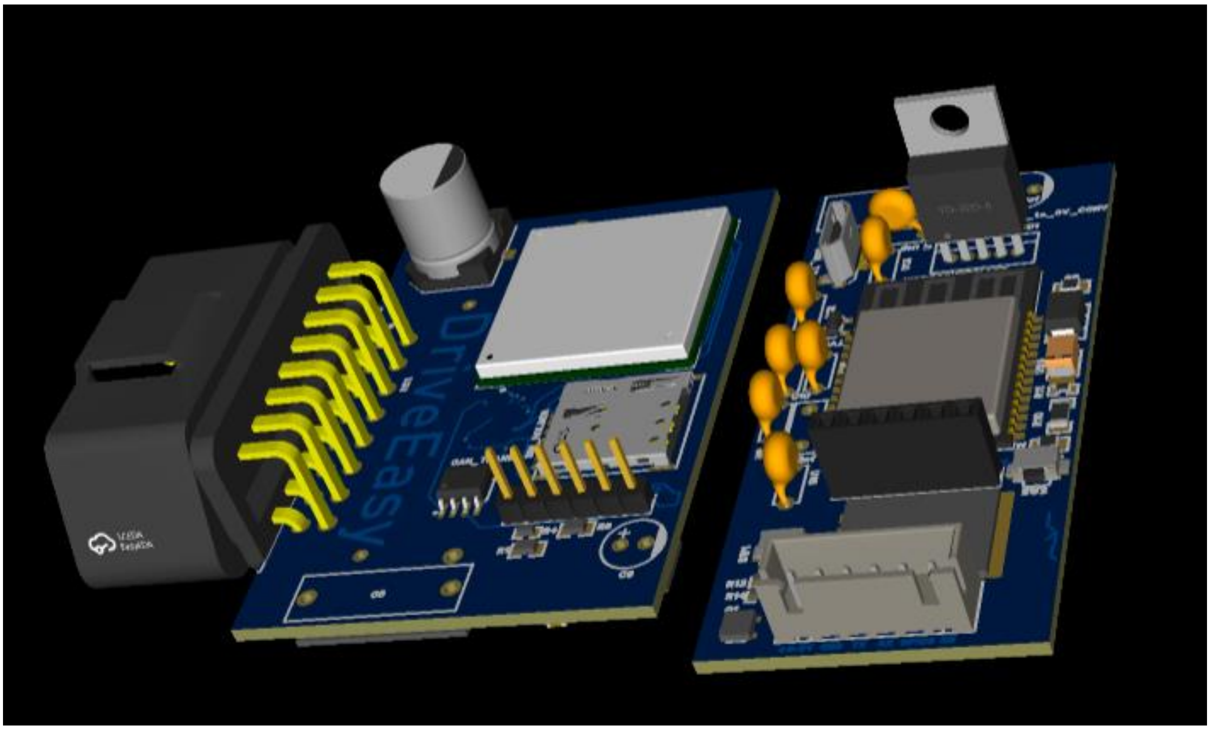
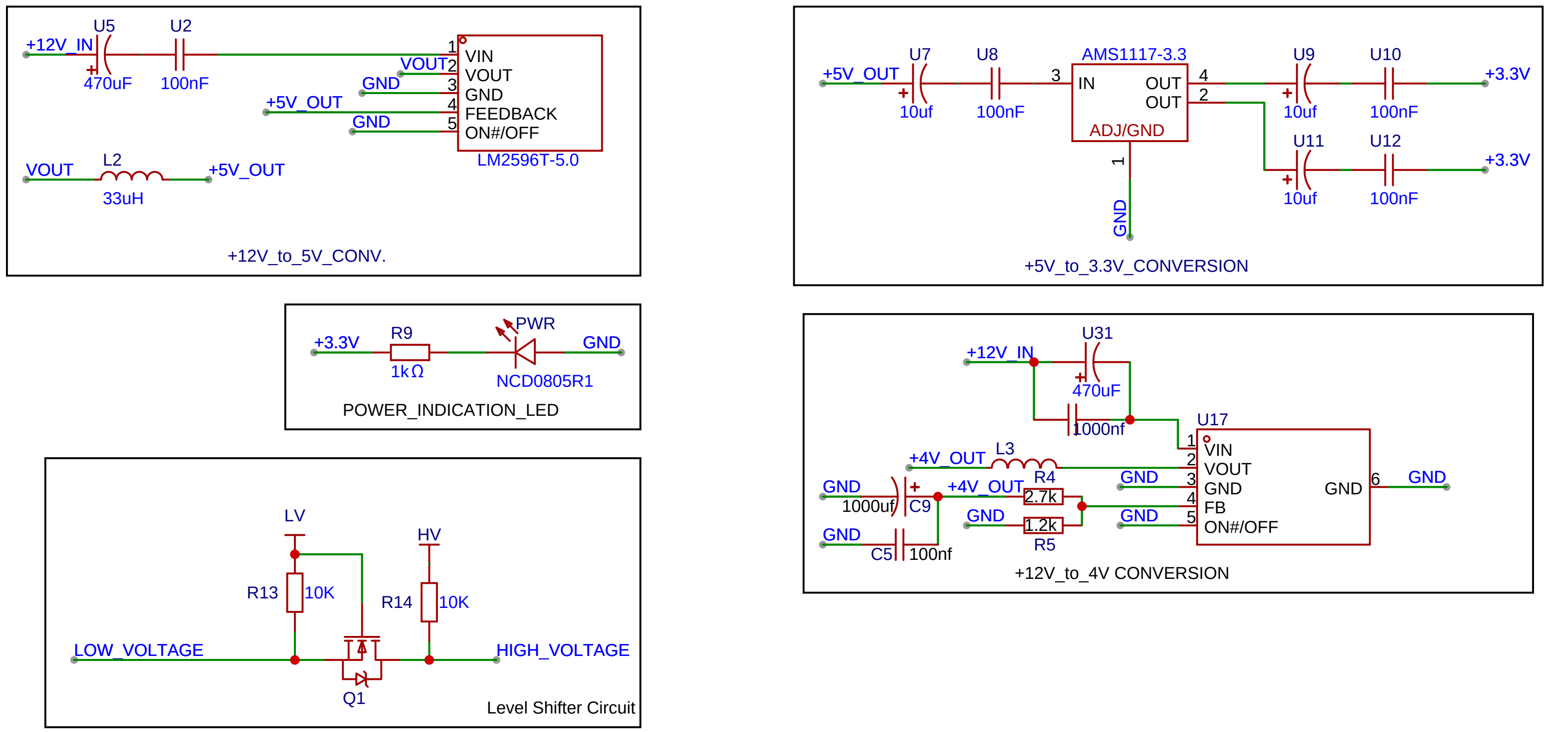


Figure : Right View



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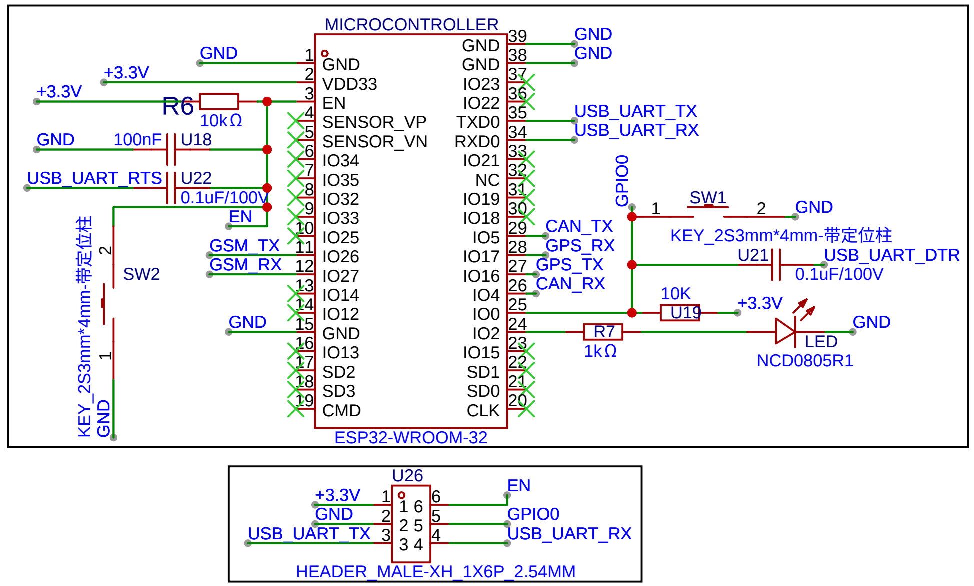

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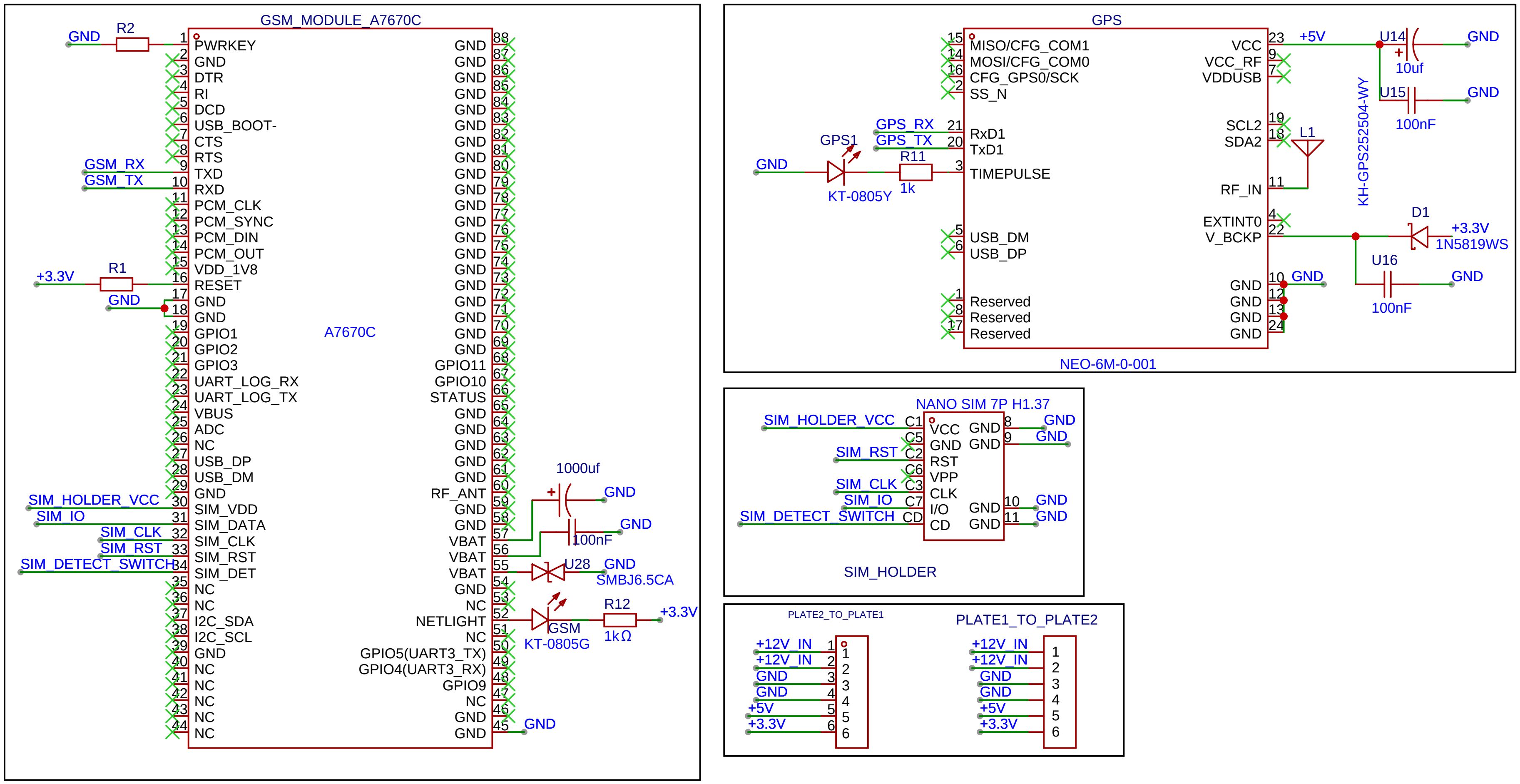

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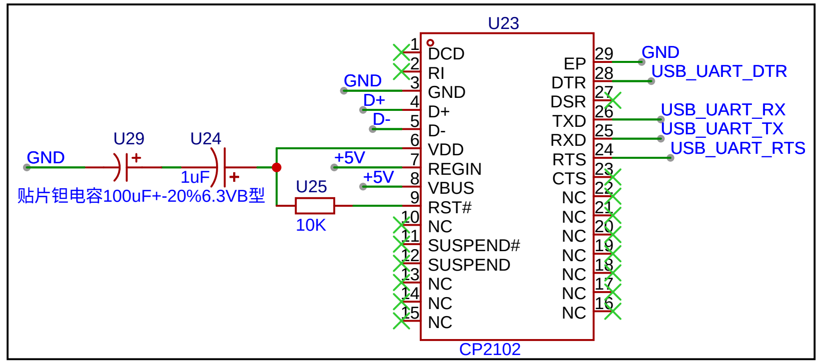

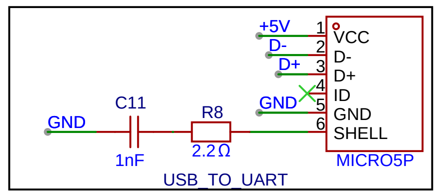

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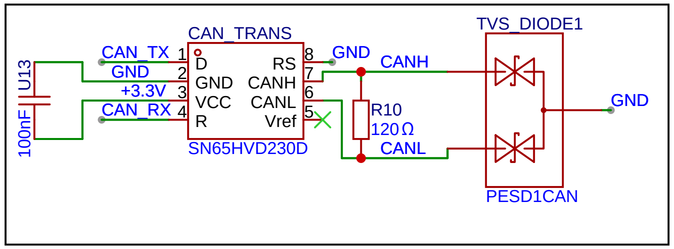

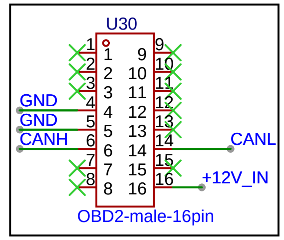

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